

MD MAHFUZUR RAHMAN

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PROFESSIONAL SUMMARY

Engineering educator and researcher with 6+ years of university-level teaching experience across engineering fundamentals, instrumentation, electronics, digital signal processing, and laboratory-centered instruction. Experienced in core undergraduate engineering courses, project-based learning, curriculum development, and student assessment. Strong background in integrating computational tools (MATLAB, Python, LabVIEW) into engineering education. Committed to inclusive, student-centered teaching and academic success for diverse and first-generation student populations. Active researcher in biomedical instrumentation and noninvasive cardiovascular sensing, with 6 peer-reviewed journal publications (4 as first author) and 2 conference proceedings.

EDUCATION

Ph.D., Mechanical & Aerospace Engineering

August 2026

Old Dominion University, Norfolk, VA | GPA: 3.87 / 4.0

- **Dissertation:** Time-Frequency Analysis of Arterial Pulse Signals of Cardiovascular Disease Patients Measured by a Microfluidic Based Tactile Sensor
- **Advisor:** Dr. Zhili Hao | NSF Grant #1936005
- **Relevant Graduate Coursework:** Engineering Mathematics; Finite Element Analysis (FEA); Engineering Optimization; Continuum Mechanics

M.S., Applied Physics, Electronics & Communication Engineering

March 2012

University of Dhaka, Bangladesh | GPA: 3.73 / 4.0 (WES)

- **Thesis:** Design of an all-optical switching structure using coupled 2D photonic crystal waveguides for optical communication
- **Advisor:** Dr. Subrata Das

B.S., Applied Physics, Electronics & Communication Engineering

September 2010

University of Dhaka, Bangladesh | GPA: 3.81 / 4.0 (WES)

- **Project:** Design of a Low-loss Y-Splitter for Optical Telecommunication using a 2D Photonics Crystal
- **Advisor:** Dr. Saeed Mahmud Ullah

TEACHING INTERESTS

- Engineering Mechanics (Statics, Dynamics, Mechanics of Materials)
- Circuits, Electronics, and Microprocessors
- Engineering Instrumentation, Measurement, and Data Acquisition
- Biomedical Instrumentation and Sensors
- Signals and Systems / Digital Signal Processing
- Engineering Mathematics
- Experimental Methods and Laboratory-Centered Instruction
- Programming-Supported Engineering Education (MATLAB, Python, LabVIEW)
- Engineering Fundamentals and First-Year Engineering
- Project-Based Learning and Undergraduate Research Mentoring
- Student Retention and Success in Undergraduate Engineering

ACADEMIC APPOINTMENTS

Graduate Teaching Assistant (Instructor)

Aug 2022 – Dec 2025

Dept. of Mechanical & Aerospace Engineering, Old Dominion University, Norfolk, VA

Graduate Teaching Assistant (Grader)

Aug 2019 – Dec 2025

Dept. of Mechanical & Aerospace Engineering, Old Dominion University, Norfolk, VA

Lecturer — Electronics & Telecommunication Engineering

Oct 2014 – Mar 2017

Atish Dipankar University of Science & Technology (ADUST), Dhaka, Bangladesh

Part-Time Lecturer

Aug 2016 – Mar 2017

Daffodil Institute of Information Technology (DIIT), Dhaka, Bangladesh

TEACHING EXPERIENCE

Graduate Teaching Assistant — Mechanical & Aerospace Engineering

Sept 2019 – Dec 2025

Old Dominion University, Norfolk, VA

Course Instructed:

- MAE 203: Materials Science Lab

Courses Supported (Grader/TA):

- MAE 111: Information Literacy and Research
- MAE 201: Materials Science
- MAE 205: Dynamics
- MAE 220: Solid Mechanics II
- MAE 434W: Project Design and Management I

Responsibilities:

- Served as primary instructor for MAE 203: Materials Science Lab — designed and delivered laboratory-centered instruction aligned with course learning outcomes in materials science and engineering fundamentals
- Implemented project-based learning activities to reinforce core engineering concepts through hands-on experimentation and data analysis
- Developed laboratory reports and performance-based assessments; assessed student performance using formative and summative methods and refined instructional strategies based on outcomes
- Coordinated laboratory content, instructional standards, and assessment practices across multiple course sections
- Supported large, multi-section undergraduate engineering courses (Dynamics, Solid Mechanics, Materials Science, Project Design) ensuring consistent grading standards and assessment practices
- Assisted in development and evaluation of exams, quizzes, and project deliverables aligned with course learning objectives
- Conducted post-exam review sessions, provided instructional continuity through guest lectures during faculty absences, and mentored student teams
- Administered end-of-semester course outcome evaluations on behalf of faculty in support of ABET accreditation documentation
- Utilized LMS platforms (Canvas/Blackboard) to manage course materials, assignments, grading, and student communication

Lecturer — Atish Dipankar University of Science & Technology

October 2014 – March 2017

Dept. of Electronics & Telecommunication Engineering (ETE), Dhaka, Bangladesh

Courses Taught:

- TPHY 108: Physics II with Lab

- ETE 208: Electrical Circuit I | ETE 209: Electrical Circuit II
- ETE 224: Electronics I | ETE 232: Electronics II | ETE 321: Electronics III
- ETE 222: Digital Logic Design
- ETE 229: Semiconductor Theory and Devices
- ETE 324: Microprocessor and Microcomputer
- ETE 331: Instrumentation and Measurement
- ETE 360: Digital Signal Processing
- ETE 401: Industrial and Power Electronics System
- ETE 499: Project/Internship

Teaching & Instruction:

- Developed and delivered introductory and core undergraduate engineering courses in circuits, electronics, instrumentation, digital logic, microprocessors, and digital signal processing, with integrated laboratory components
- Designed outcome-driven syllabi, lecture materials, assignments, laboratory experiments, and examinations aligned with program learning objectives
- Integrated MATLAB-based signal analysis, numerical methods, and data visualization into electronics, instrumentation, and signal processing courses to support engineering problem-solving and experimental interpretation
- Applied project-based learning in laboratory and design-oriented courses to strengthen conceptual understanding and hands-on skills
- Implemented rubric-based grading and structured assessment methods to evaluate student learning outcomes and course effectiveness

Student Mentoring & Curriculum Development:

- Supervised undergraduate final-year projects and internships, guiding students through project planning, implementation, documentation, and technical presentation
- Provided academic mentoring and formative feedback to support students from diverse educational backgrounds
- Served on the department curriculum development committee, contributing to course sequencing, content alignment, and instructional consistency across multiple sections

Part-Time Lecturer — Daffodil Institute of Information Technology

August 2016 – March 2017

Dept. of Computer Science and Engineering (CSE), Dhaka, Bangladesh

Courses Taught:

- CSE 123: Introduction to Electrical Engineering
- CSE 124: Introduction to Electrical Engineering Practical
- CSE 438: Microprocessor and Assembly Language

Responsibilities:

- Taught introductory engineering and computing courses with integrated laboratory components
- Delivered lectures and lab sessions in electrical engineering fundamentals and microprocessor systems
- Designed assessments and provided structured feedback to support student learning

STUDENT MENTORING

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| • Md. Rajon Ahmed — Undergraduate Final-Year Internship
<i>Dept. of ETE, ADUST, Dhaka, Bangladesh</i> | Spring 2016 |
| • Md. Kamrul Hasan — Undergraduate Final-Year Internship
<i>Dept. of ETE, ADUST, Dhaka, Bangladesh</i> | Spring 2016 |
| • Md. Tauhidur Rahman — Undergraduate Final-Year Project
<i>Dept. of ETE, ADUST, Dhaka, Bangladesh</i> | Summer 2015 |
| • Muhammad Ali — Undergraduate Final-Year Project
<i>Dept. of ETE, ADUST, Dhaka, Bangladesh</i> | Summer 2015 |
| • Homayun Kabir — Undergraduate Final-Year Internship
<i>Dept. of ETE, ADUST, Dhaka, Bangladesh</i> | Summer 2015 |

SKILLS SUMMARY

- **Programming & Computation:** MATLAB, Python, LabVIEW, SPSS; Machine Learning (graduate coursework - ECE 607)
- **Signal Processing & Time-Series Analysis:**
 - **Spectral & Time-Frequency Analysis:** Fast Fourier Transform (FFT); Continuous Wavelet Transform (CWT); Complex Demodulation (CDM)
 - **Filter Design & Denoising:** FIR (Kaiser Window, Parks-McClellan/Equiripple); IIR (Butterworth); Adaptive Filtering; Discrete Wavelet Transform (DWT, Multilevel Decomposition)
 - **Adaptive Decomposition:** Empirical Mode Decomposition (EMD); Hilbert-Huang Transform (HHT); Hilbert Vibrational Decomposition (HVD)
 - **Developed/Novel Methods (Ph.D. Research):** Hybrid masked based EMD-HVD, and hybrid CDM-HVD framework for instantaneous parameter estimation in nonstationary physiological signals
- **Engineering & Design Tools:** AutoCAD, Autodesk Inventor, KiCad, Multisim
- **Data Acquisition:** LabVIEW-based DAQ (NI DAQ hardware), real-time signal monitoring and processing
- **Clinical Research Tools:** REDCap (data collection, entry, and quality management)
- **Laboratory & Instruction:** Experimental design, lab safety, hands-on instrumentation, rubric-based assessment
- **Learning Management:** Canvas, Blackboard; digital course delivery; learning-outcome-aligned assessment design

RESEARCH

Research Areas:

- Biomedical instrumentation and noninvasive cardiovascular measurement
- Time-frequency signal analysis and motion artifact characterization in physiological signals
- Soft polymer microfluidic tactile sensor design, fabrication, and clinical validation

Research Output:

- 6 peer-reviewed journal publications (4 as first author); 2 conference proceedings (both as first author); conducting NSF-funded research (Grant #1936005).

PUBLICATIONS

Preprints

2026

- **Rahman, M.M.**, Hasan, M., May, J. F., Herre, J. M., Reynolds, L. J., & Hao, Z. (2026). Subject-Specific Cardiovascular Signatures Revealed by Arterial Pulse Signals in Patients with Heart Disease Part I: General Cases. [doi:10.20944/preprints202608.2302.v1](https://doi.org/10.20944/preprints202608.2302.v1)
- Hasan, M., **Rahman, M.M.**, May, J. F., Herre, J. M., Reynolds, L., & Hao, Z. (2026). Subject-Specific Cardiovascular Signatures Revealed by Arterial Pulse Signals in Patients With Heart Diseases — Part II: Special Cases. [doi:10.20944/preprints202608.2324.v1](https://doi.org/10.20944/preprints202608.2324.v1)

Journal Articles (Peer-Reviewed)

2026

- **Rahman, M.M.**, Hasan, M., May, J.F., Herre, J.M., Reynolds, L., & Hao, Z. (in press). Radial and carotid arterial pulse signals for assessing cardiovascular function at rest and during post-exercise recovery in a heart transplant patient: a case study. **Medical Sensors & Imaging** (IOP Publishing) [[Accepted](#); [doi:10.1088/2977-8425/ae4b40](https://doi.org/10.1088/2977-8425/ae4b40)]

2025

- **Rahman, M.M.**, et al. (2025). An Analytical Model of Motion Artifacts in a Measured Arterial Pulse Signal—

Part I: Accelerometers and PPG Sensors. **Sensors**, 25(18), 5710. doi:10.3390/s25185710

- **Rahman, M.M.**, et al. (2025). An Analytical Model of Motion Artifacts in a Measured Arterial Pulse Signal—Part II: Tactile Sensors. **Sensors**, 25(18), 5700. doi:10.3390/s25185700
- **Rahman, M.M.**, et al. (2025). Motion Artifacts (MA) At-Rest in Measured Arterial Pulse Signals: Time-Varying Amplitude in Each Harmonic and Non-Flat Harmonic-MA-Coupled Baseline. **Biosensors**, 15(9), 578. doi:10.3390/bios15090578
- Toraskar, S., **Rahman, M.M.**, & Hao, Z. (2026). Pulse Measurement Alters the True Pulse Signal in an Artery: A Coupled String-SDOF Model. **ASME J. Engineering and Science in Medical Diagnostics and Therapy**, 9(1), 011207. doi:10.1115/1.4069099

2023

- Hao, Z., **Rahman, M.M.**, et al. (2023). Axial Wall Displacement at the Common Carotid Artery is Associated With the Lamb Waves. **ASME J. Engineering and Science in Medical Diagnostics and Therapy**, 6(1), 011008. doi:10.1115/1.4056267

Conference Proceedings

2021

- **Rahman, M.M.**, et al. (2021). Measurement of Post-Exercise Response of Local Arterial Parameters Using an Adjustable Microfluidic Tactile Sensor. **IEEE EMBC 2021**, pp. 1284–1287. doi:10.1109/EMBC46164.2021.9630432

2020

- **Rahman, M.M.**, et al. (2020). Improved Vibration-Model-Based Analysis for Estimation of Arterial Parameters From Noninvasively Measured Arterial Pulse Signals. **ASME IMECE 2020**, V005T05A079. doi:10.1115/IMECE2020-24551

HONORS & AWARDS

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| • Tiwari Endowed Graduate Scholarship, Old Dominion University | 2021 |
| • National Science & ICT (NSICT) Fellowship, Bangladesh | 2010 |
| • Merit-Based Scholarship, University of Dhaka | 2008 |

CERTIFICATIONS

- Biomedical Research — Basic, CITI Program
Issued March 2026 · Expires March 2029 · Credential ID: 72079387 (*Active*)
- Biomedical Research — Refresher 2, CITI Program
Issued December 2023 · Expired December 2025 · Credential ID: 57666724
- Biomedical Research — Refresher 1, CITI Program
Issued November 2021 · Expired November 2023 · Credential ID: 43904237
- Biomedical Responsible Conduct of Research, CITI Program
Issued October 2020 · Credential ID: 39136829

PROFESSIONAL & COMMUNITY SERVICE

- **Conference Reviewer:** International Conference on Imaging, Vision & Pattern Recognition (IVPR)
- **Volunteer:** Virginia Medical Reserve Corps (MRC) May 2020–Present
Volunteered in community public-health initiatives including COVID-19 vaccination clinics, community-based health screening programs, patient intake, and health event coordination

INDUSTRY EXPERIENCE

Manager — Merchandising & Production

Apr 2017 – Aug 2019

Afiya Knitwear Ltd., Savar, Bangladesh

- Directed end-to-end merchandising strategy and execution, from buyer negotiation and costing to production planning and shipment delivery
- Led cross-functional teams to ensure on-time delivery, cost control, and adherence to buyer quality and compliance standards
- Oversaw sourcing, sampling approvals, and risk mitigation for production delays or quality issues

REFERENCES

Available upon request.